Lean proof simplification for physical safety and same data packing

A sixth review with every earlier proposal assumed available

Source commit `9d9d1123b39927d30363f1dcd53c23f0575c30ee`

1 Findings and recommendation

The sixth iteration is three small proof-only deletions in newly available clients. The main recommendation removes 15 physical lines across three complete modules, introduces no helper or import, and preserves every public statement and data definition. Compile these bounded changes separately before adoption. A further two-line disjoint-mask cleanup is included as secondary housekeeping, not as another substantive idea.

Candidate	Full-file lines	What disappears
Actual controller positions	178 → 174; −4	Nonlinear detours
Actual rank lower bound	138 → 133; −5	Intermediate square bound
Same-data rephasing	122 → 116; −6	Local function equality
Main recommendation	438 → 423; −15	Zero helper cost
Disjoint-sum housekeeping	57 → 55; −2	Named rewrite equality
All four files	495 → 478; −17	Five proof-body edits

The source has moved beyond the previous open boundary. It already contains actual controller positions, a one-dummy ordinary no-wrap theorem under physical position bounds, exact packing population and rank bounds, and source-preserving rephasing. These constructions are baseline, not discoveries of this report. The new work is only the residual proof deletion after those results and all prior proposal contracts are assumed available.

The three changes have different mathematical reasons. Fixed actual coefficients turn position inequalities into linear integer arithmetic. The rank comparison follows directly from $2u \le 3u < 2^{3u}$. Rephasing reuses the exact recurrence and decomposition fields by rewriting their rotation inputs, while leaving the distinct scalar transport proof intact. Each is explained against its unchanged domain and witnesses below.

All new Lean candidates are UNCOMPILED. No Lean, Lake or Elan executable is available in this task environment, and no toolchain was installed. Static audits, exact patches, source inspection and finite arithmetic checks do not establish elaboration, kernel acceptance, axiom equality or a speed improvement. The PDF is compiled separately; its successful build says nothing about Lean acceptance.

The pinned active Lean tree contains 702 files and 87,415 physical lines. The final proposal baseline contains both excluded collections and 43 placed manuscripts, including the delivered fifth report. Three tempting optional-support proof deletions were withdrawn after the final comparison. Novelty is assessed against their useful contracts, not merely matching filenames. This report does not claim a new general theorem or completion of the full polynomial–Con(ZFC) equivalence.

2 Start after every earlier proposal

Fixed source and final comparison. Production source remains pinned at `9d9d1123b399`; the preceding exclusion cutoff was `91dce223cda8`. At the source pin there were 36 placed manuscripts and five incoming ZIPs, one byte-identical to the fifth delivery. The final bounded comparison at `60fcb2a69b7f` contains 43 placed manuscripts and no incoming ZIPs: the five archives were relocated and two new manuscripts appeared. The original 36 trees, all four candidate target files and 15 selected dependency/context files are unchanged. This is not an assertion about the entire import closure.

Assumed applied means semantic availability, not a compiled super-patch. Every useful earlier outcome counts, including optional, deferred, uncompiled and competing designs. New names do not make an old argument new. The supplied `catalog/` records corpus identities, semantic exclusions and review receipts, not duplicated original ZIPs or prior PDFs.

Four incoming families at the source pin	Outcomes already granted
New obligations	Shared Jones finalizer facts; affine carrier soundness; homogeneous-star enumeration; fixed-power dependencies
Proof simplifications	Gamma certificate and lifting; arbitrary-exponent LTE; reset-separated carry locality; low-field and subset options
Additional certificates	Common-modulus literal-K evaluation; fixed-power propagation; weighted point update; reflected-source boundary; batch and affine-fiber directions
Other round-five report	Sidon pair cleanup; order-zero wrappers; definitional iterate depth; terminal-field identity direction

The final two manuscripts materially affect the result. Packet zp adds all-offset reflection, prime-support coprimality, arbitrary-cardinality seed translation and universal LTE/order organization. Packet zq adds restricted subset sums, Boolean-free affine coverage, optional-support retirement, physical-composition directions, whole-Gamma residue and LTE companions. Their notes-only alternatives are excluded too.

In particular, zq's optional-support retirement note explicitly proposes retiring the `optional_next_lt` and `optional_stride_lt` clients. This report also conservatively withdraws `position_ne_three`, whose production consumers serve that same cell-three path. Those three cuts would save nine lines at the frozen source, but are not new post-adoption savings. The final main recommendation is therefore 15 lines, not 24.

Closest retained precedents. E41's guarded predecessor normalization $a^t = a\,a^{t-1}$ does not supply these subtraction-free budget and injectivity deletions. Old zc/zf threshold methods and `Nat.lt_two_pow_self` rank techniques remain granted; only their exact new-client residual is counted. Coherent-carry views, uniqueness, reset locality and modular projections remain baseline; rephasing contributes only a two-field local rewrite. Every fifth-delivery outcome, including withdrawn `q_prefix`, remains excluded.

Upstream progress is not new credit. The source-pin adoption ledger reports six earlier families, 81 net lines and 3097 cumulative lines. Four more prior-work adoptions at the final cutoff report 156 additional lines, raising that total to 3253. These are documentary upstream claims, not reruns or this review's savings. The preceding cutoff-to-source interval has 17 graph commits, 34 changed repository files and 17 active Lean files (+1066/ − 144 lines); merge commits are not independent changes. Finite-support mutation composition, actual rank-congruence selection, rank-forced temporal exponentiation, remaining positive Pell/source coordinates, the full converse and internal-ZFC equivalence are not established by these local deletions.

3 Make the actual position arithmetic linear

Use the fixed coefficient already present in the statement. In `CurrentControllerPositions2011.lean:13–17`, write

$$p = 120199, \qquad A = p^k, \qquad P_k(j) = 2 + (p-1)j.\,\mathrm{val}, \qquad j : \mathrm{ZMod}(p^k).$$

The ordinary cell modulus is $p^{k+1} = pA$. These are the actual controller positions, not an arbitrary affine family. Unfolding only `position`, `auxiliary` and one `pow_succ` turns the local obligations into Presburger arithmetic with coefficient 120198. The term p^k can remain an opaque natural atom. Only the small actual auxiliary 120199 is exposed, not the huge block base or literal kernel K. No symbolic power is evaluated as an enormous numeral, and no generic theorem is added.

Sharp budget. The canonical representative satisfies $j.\,\mathrm{val} + 1 \le A$. Therefore

$$P_k(j) + A + p = 2 + (p-1)j.\,\mathrm{val} + A + p \le 2 + (p-1)(A-1) + A + p = pA + 3.$$

The candidate replaces a multiplied inequality, a separate power equality and nonlinear arithmetic by the following entire body:

```
have hv : j.val+1 ≤ auxiliary^k := ZMod.val_lt j
simp only [position, pow_succ, auxiliary] at hv ⊢
omega
```

This preserves the subtraction-free public budget, hence its existing natural-subtraction endpoint and safe-next-cell clients. The source is `CurrentControllerPositions2011.lean:24–43`.

Injectivity needs only cancellation by a fixed positive integer. If $P_k(i) = P_k(j)$, then

$$2 + 120198\,i.\,\mathrm{val} = 2 + 120198\,j.\,\mathrm{val} \quad \Longrightarrow \quad i.\,\mathrm{val} = j.\,\mathrm{val}\,.$$

The candidate applies `ZMod.val_injective`, unfolds the two fixed definitions in the equality, and lets `omega` cancel the coefficient. This replaces a separate positivity fact and `nlinarith`; no primality argument or new finite-order principle is needed. See `CurrentControllerPositions2011.lean:79–84`.

Stop before the prior optional-support design. The final exclusion comparison revealed an earlier direction to retire the optional-support API. Accordingly, the initially considered deletions in `optional_next_lt`, `optional_stride_lt` and `position_ne_three` are withdrawn, even though they are valid-looking local arithmetic simplifications. They receive neither a retained patch nor savings credit. Only the budget and injectivity clients survive the assumed-applied comparison.

The actual positions, cyclic embedding, support and optional-cell definitions remain unchanged in the candidate. Its existing safety and cardinality clients are retained verbatim. Both changed statements still cover $k = 0$; no $r_w < k$ assumption is added to them. The physical guard and same-word consequences on the next page are also unchanged source results, not additional cuts.

Cost and scope. Two proof bodies shrink from 11 to 7 physical lines. The complete module shrinks from 178 to 174 lines and from 7345 to 7173 UTF-8 bytes; all helpers are included and no new helper or import is introduced. This is a fresh-client cleanup, not a new affine geometry or support-cardinality construction. It does not claim that `omega` compiles faster than `nlinarith`.

4 Keep the existing physical no wrap theorem

The one-dummy no-wrap obligation is already discharged at this pin. It is not a new missing theorem. The previous page shortens the position arithmetic that feeds the safety interface; it does not replace or take credit for the safety construction itself.

Let $\rho = 2^{47}$, $m = 2011N$, $W = 47 \cdot 2011s$, and

$$D_j = \rho^{977+2011j.\,\mathrm{val}}, \qquad \Delta = (K + 2^W)D_j.$$

The literal exponents are at most 2020 and the total coefficient mass is $7659829036801887 < \rho^2$, so $K < \rho^{2022}$. Thus

$$KD_j < \rho^{2999+2011j.\,\mathrm{val}}, \qquad 2^W D_j = \rho^{977+2011(j.\,\mathrm{val}+s)}.$$

See `CurrentDummySafety2011.lean:22–72`. The canonical-position hypotheses are

$$j.\,\mathrm{val}+1 < N, \qquad j.\,\mathrm{val}+s < N.$$

They imply $m \geq 4022$ and leave far more than the two high fields required by `safe_increment_bound`: $0 < \Delta < \rho^{m-2}$. They allow $s = 0$, but exclude $N = 1$ and a last-cell update. Dropping either condition is not an equivalent simplification. See `CurrentDummySafety2011.lean:74–119`.

Why the ordinary bound follows. The actual original normalized word F satisfies $F < \rho^{m-1}$. Set $Q = \rho^{m-2}$. Since $m \geq 2$, $\rho \geq 2$, and both inequalities are strict natural bounds,

$$F + \Delta + 2 \leq Q(\rho + 1) < Q\rho^2 = \rho^m.$$

The middle inequality uses $Q > 0$ and $\rho + 1 < \rho^2$. Therefore

$$F + \Delta < \rho^m - 1.$$

This is an ordinary inequality, not merely a congruence. It is the premise required to turn normalized transport into the exact natural increment. The source's generic `nonwrap_of_short` and actual `normalized_nonwrap_of_safe` already implement this argument; see `CurrentDummySafety2011.lean:121–150`.

The same data must remain attached. With the original a, dummy and marker functions, Boolean bounds, a safe unused cell with $\mathrm{dummy}(j) = 0$, and the old normalization facts, the existing construction obtains a new normalization for the updated dummy function. It preserves the ordinary source increment $C' = C + D_j$ and the ordinary normalized increment $F' = F + \Delta$. Its rank equality is then about those same old and new words, carries and digits, not a fresh unrelated word. See `CurrentDummySafety2011.lean:152–211`.

What is deliberately not proposed. Replacing the explicit physical guards by a new abstract safe-position record would add an already familiar interface, not reduce the residual proof. Deriving only a modular update would weaken the ordinary equality. Generalizing the fixed coefficient arithmetic to a parameterized base would reintroduce positivity and multiplication lemmas. The existing small safety proof is kept unchanged, including its no-high-mask-premise design.

5 Use the exponential rank bound at its final exponent

Replace an intermediate square estimate by one standard inequality. Set $u = \mathrm{width} \cdot N$ and $q = 2^u$. The fresh client `rank_gt_twice_window` currently derives $u < q$, $q \geq 2$, and $q^2 \geq 2$ before proving $2u < q^3$. Instead use

$$2u \leq 3u < 2^{3u} = (2^u)^3 = q^3.$$

The strict step is the existing `Nat.lt_two_pow_self` at exponent $3u$. It holds even when $u = 0$; there is no need to manufacture positivity merely to prove this numerical comparison. The replacement is:

```
have hc : 2*(width*N) < q N^3 := by
  rw [q_binary, ← pow_mul]
  exact (show 2*(width*N) ≤ (width*N)*3 by omega).trans_lt
    Nat.lt_two_pow_self
```

The printed line break is for readability; the delivered candidate has three physical lines in place of eight. The public statement retains [NeZero N], $C > 0$ and $C + F < q$, since the rest of the rank construction still uses its original domains. See `CurrentPackingRank2011.lean:77–100`.

Do not bypass the natural representative. The rank is the existing integer `windowRank` with fourth source port `highPort+beta`, shifted exactly once. Its natural representative is

$$\begin{aligned} B = {} & (C-1) + (LJ+1) + q\big(F + (HJ-1)\big) \\ & + q^2(q-C) + q^3(q-F-1), \end{aligned}$$

where $L = \mathrm{lowPort}$, $H = \mathrm{highPort}$ and J is the repunit. From $C > 0$ and $C + F < q$, the top complement satisfies $q - F - 1 \geq 1$, so $q^3 \leq B$. The unchanged cast theorem checks every natural subtraction before identifying $(B : \mathbb{Z})$ with the actual rank. Composing these facts yields $2u <$ rank without changing B, the source, the normalized word or the mask ports. See `CurrentPackingRank2011.lean:22–75`.

The exact population construction stays intact. For $A, F, M, H < 2^u$ and disjoint masks $A \,\&\, M = F \,\&\, H = 0$, no addition carry crosses a window. The three lower blocks are strict representatives; the top block need not be restricted. Binary population σ therefore splits across four blocks. The disjoint-sum identities and complement identities give

$$\sigma\big((A+M) + 2^u(F+H) + 2^{2u}(2^u - 1 - A) + 2^{3u}(2^u - 1 - F)\big) = 2u + \sigma(M) + \sigma(H).$$

Calibration $\sigma(M) + \sigma(H) = u + 2$ yields $3u + 2$. The actual client still supplies the repeated-mask calibration and the same B witness; no Pell population premise is introduced. See `NativePackingPopulation.lean:19–55` and `CurrentPackingRank2011.lean:102–136`.

A secondary two-line deletion. In `disjoint_add_lt`, directly rewrite the goal with `JM1984.add_eq_land_add_lor`, disjointness and `zero_add`; then retain `Nat.or_lt_two_pow`. This removes a named intermediate equality, not a new population argument. The two complete files shrink 138 → 133 and 57 → 55 lines, seven lines total with zero helper or import cost. Exponential domination and general rank/cast interfaces were already excluded; only these residual client steps are credited.

6 Rewrite the two normalization fields directly

Delete a local function equality that only serves two fields. The source theorem `normalization_of_rotation_eq` transfers the same carry and digit functions across an equality of actual binary rotations. For any $n > 0$, source function $c : \mathrm{ZMod}(n) \to \mathbb{N}$, and natural shifts W, W', assume

$$\mathrm{rotation}_{47}(W', c) = \mathrm{rotation}_{47}(W, c).$$

The raw recurrence input is a fixed cyclic convolution plus that rotation. Consequently the recurrence and decomposition fields can each be reused after one exact rewrite. The candidate replaces the auxiliary function-extensionality proof and both subsequent `rw` steps with:

```
refine ⟨hf.carry_bound,hf.digit_bound,?_,?_,hf.word_bound,?_⟩
· intro i
  simpa only [CurrentCarryConstruction2011.raw,hr]
    using hf.recurrence i
· intro i
  simpa only [CurrentCarryConstruction2011.raw,hr]
    using hf.decomposition i
```

These are restricted simplifications at the two exact consumers; the same raw definition is unfolded twice instead of once in a function equality. This does not claim a new normalization-uniqueness theorem. The source range is `CurrentWordRephase2011.lean:23–47`.

The scalar transport proof is intentionally retained. Put $M = \rho^n - 1$. Equality of rotations and the existing `word_binaryRotation` congruences give

$$2^{W'} \,\mathrm{word}(c) \equiv 2^{W} \,\mathrm{word}(c) \pmod{M}.$$

Adding $K\,\mathrm{word}(c)$ and composing with the old transport establishes the new transport congruence. The ordinary multipliers $K + 2^{W'}$ and $K + 2^{W}$ need not be equal; the proof never identifies them. Keeping this field derivation is necessary even though the recurrence data is unchanged.

Downstream rephasing retains the particular witnesses. The actual stride condition $(s : \mathrm{ZMod}\, N) = (t : \mathrm{ZMod}\, N)$ gives equal markers, equal complete source functions and equal physical rotations. The existing `normalization_rephase` reuses the same carries and digit function. The packing theorem then recalculates the positive scalar quotient for the new ordinary multiplier while retaining those functions, the same tiles and the same dummy data. See `CurrentWordRephase2011.lean:51–120`.

The generic theorem changed here has no Boolean-source assumption; its source function remains an arbitrary natural-valued function. Booleanity is used only by the existing constructed-word client to establish rotation equality. No new normalization is chosen, no rank-growth or rank-congruence hypothesis is inserted, and no $s, t < N$ restriction is introduced.

Cost and risk. The whole module shrinks from 122 to 116 physical lines and from 6023 to 5925 bytes; no helper, import, public declaration or theorem type changes. The exact remaining elaboration risk is the orientation and definitional unfolding of the two `simpa only` calls. Generic source views, carry uniqueness, modular projections and same-data reset interfaces were already available and are not counted as new work.

7 Charge full source cost and preserve the contracts

Count complete production files, not just shortened proof snippets. The main batch changes two position proof bodies, one rank proof body and one rephasing proof body. The secondary mask cleanup changes one more. All four files together shrink from 495 to 478 physical lines. There are no new production helpers, imports, definitions or structures; the total therefore needs no hidden helper allowance. Supplied check scripts, source snapshots and report text are evidence, not reductions in the maintained Lean source.

Source module	Measured change	Invariant to retain
`CurrentControllerPositions2011`	178 → 174 lines; 7345 → 7173 bytes	Same p, positions, optional cell and support
`CurrentPackingRank2011`	138 → 133 lines; 6099 → 5843 bytes	Same integer rank and natural representative
`CurrentWordRephase2011`	122 → 116 lines; 6023 → 5925 bytes	Same carries, digits and scalar proof
`NativePackingPopulation`	57 → 55 lines; 2168 → 2137 bytes	Same strict bounds and disjointness

The savings are source facts, not performance measurements. Replacing `nlinarith` by `omega`, or a named equality by `simpa only`, may change elaboration behavior. No timing, heartbeat reduction, memory saving or proof-term size reduction has been measured. These candidates are worth trying because their proof dependencies and mathematical steps are simpler at the exact client, not because text length predicts runtime.

Independent arithmetic evidence is deliberately limited.

Check family	Recorded finite scope
Actual-position formulas	141,460 affine points; 2,940 injectivity pairs; unchanged-baseline regressions include 250 support sets and 352 safe-exponent triples
Population and rank models	66,430 exact-population cases; 2,340 unrestricted-top cases; 13,869 cast and strict-rank cases
Same-data rephasing model	32,832 reduced-radix cases; 242,568 recurrence and decomposition instances; 30,916 cases with changed quotient

The population and rephasing models do not evaluate the complete literal 2011/47 construction. They test the arithmetic contracts and their edge cases, not universal quantifiers or Lean syntax. The models include $u = 0$, unrestricted top blocks, non-Boolean source digits, and one-cell cyclic wrap where applicable; they do not enlarge the public hypotheses.

Negative controls reject missing disjointness or strict block bounds, unsafe final/temporal positions, a nonperiodic rephase, and identifying ordinary multipliers across equal rotations. One reduced-radix example has $\rho = 4$, $n = 1$, $K = C = 1$, $F = 2$, and the same carry 0: rephasing $W = 0$ to $W' = 2$ keeps the rotation and words but changes 2^W from 1 to 4 and the positive quotient from 1 to 2. This illustrates why the retained transport and quotient logic cannot be deleted.

8 Validate one dependency ready module at a time

Lean compilation remains the adoption gate. The packaged checks reconstruct the exact candidate patches, compare pinned source hashes and public headers, and check that non-proof code and contexts are unchanged. The integrated audit replayed all six main-batch and 24 optional-inclusive file-patch orders, and rejected 15 source or obsolete-patch mutations in both normal and optimized Python. Finite arithmetic is independent corroboration only. The candidate files must still elaborate and pass the project's original checks.

Use the declared Lean `v4.32.0` and Mathlib revision `81a5d257c8e410db227a6665ed08f64fea08e997`. Begin in a clean worktree at the source pin with its dependencies available and record a successful unmodified baseline. Installation and cache retrieval are separate environment actions; neither was performed in this review.

Do not rely on `LAKE_JOBS=1`. The repository's corrected build warning says it does not limit the worker count. Run one Lean process/module per invocation, and only after that module's dependencies are compiled. Rebuild every changed main module before its Checks module. Use fully qualified `DiophantineConsistency.` target names. Do not launch an aggregate target over stale dependencies and never set `LEAN_NUM_THREADS=0`.

1. Apply `positions_combined.patch` alone for both bodies; the two isolated patches are alternative single edits against the original baseline, not a sequential patch series. Build `CurrentControllerPositions2011`, then its Checks module. Inspect the $k = 0$ cases, exact canonical values, optional cell 3, support cardinality and original $r_w < k$ guard. Build the unchanged safety main and its Checks module when their dependencies are ready; verify that the existing physical-guard no-wrap and ordinary increment statements remain intact.

2. Apply the rank patch alone. Build `CurrentPackingRank2011`, then `CurrentPackingRank2011Checks`. Confirm the type of `Nat.lt_two_pow_self` after rewriting q^3, the natural-to-integer cast, the unchanged $C > 0$ and $C + F < q$ guards, and the once-shifted high-mask port.

3. If the secondary cleanup is wanted, apply it to `NativePackingPopulation`; rebuild that main, then its Checks module, followed by the rank main and Checks. Do not build the rank against a stale population dependency.

4. Apply the rephasing patch. Build `CurrentWordRephase2011`, then its Checks module. Confirm both restricted field rewrites, unchanged scalar congruence, and the same carry/digit functions in the packing endpoint. Test a nonzero whole-period stride difference, not only $s = t$.

5. Only after every prerequisite is compiled, run the encompassing target and original downstream checks. Inspect the final diff; record exact versions, invocations, exit codes and failures. If required for adoption, compare `#print axioms` on every changed public theorem and dependent endpoint before and after. This packet does not claim such a comparison has run.

Reproducible evidence. The archive contains exact before/after sources and patch files under `safety/`, `population/` and `rephase/`; their manifests and read-only checkers record full-file costs and finite scopes. `catalog/` records the exclusion corpus, earlier delivery and source history. `audit/` provides the independent integrated checker. The report's pinned source links identify exact theorem ranges rather than a moving branch. `report/build.sh` rebuilds this PDF from the supplied `report.tex` and `inputs/` using an installed TeX Live, without downloads.